NNN-E-920a
February 19, 1963
SUPERSEDING
Int. Fed. Spec. NNN-E-00920 (ARMY-CmlC)
June 30, 1961, and
Fed. Spec. GG-E-951a
September 22, 1953

FEDERAL SPECIFICATION

EXTRACTION ASSEMBLY, WITH FRIEDRICHS CONDENSER

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers an extraction assembly consisting of a Friedrichs condenser, a Soxhlet extractor, and a boiling flask for laboratory use.

1.2 Classification.

1.2.1 *Size and design.* The condensers, extractors, and flasks shall be of the size and design, as specified (see 6.2).

1.2.1 *Friedrichs condenser.*

Small
Medium
Large

1.2.2 *Soxhlet extractor.*

Small
Medium
Large

1.2.3 *Boiling flask.*

Small
Medium
Large

Design A—flat bottom
Design B—round bottom

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

Federal Specifications:

DD-G-541—Glass (Laboratory).
PPP-B-601—Boxes, Wood, Cleated-Plywood.
PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636—Box, Fiberboard.
PPP-C-843—Cushioning Material, Cellulosic.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Adminis-

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tration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Uniform Freight Classification Rules.

(*Application for copies should be addressed to* Uniform Classification Committee, 202 Union Station, Chicago 6, Illinois.)

Commercial Standard CS-21—Interchangeable Taper-Ground Joints, Stopcocks, Stoppers, and Spherical-Ground Joints.

(Copies may be obtained from Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.)

3. REQUIREMENTS

3.1 **Material.** The components of the extraction assembly shall be made of borosilicate glass conforming to type I, class A of DD-G-541.

3.1.1 *Ground glass joints.* The standard taper-ground glass joints shall meet the dimensional requirements of Commercial Standard CS-21.

3.2 **Style and design.** Extraction assembly shall consist of a Friedrichs condenser, a Soxhlet extractor, and a boiling flask.

3.2.1 *Friedrichs condenser.* The condenser shall be of the Friedrichs type and shall have a standard taper-ground inner bottom joint as follows: Small-35/45, medium-45/50, large-55/50, also on inlet and outlet water connection at the top, and a side tube. Side tube shall be tooled to take a No. 3 rubber stopper, and shall be joined to the condenser jacket at a 75° ± 2° angle, measured from the top. Overall length, including the drip tip at the bottom shall be 335 ± 10 millimeters (mm.). Inner tube length shall be 190 ± 3 mm. Length of side tube shall be 65 ± 3 mm.

3.2.2 *Soxhlet extractor.* The extractor shall be of the Soxhlet type consisting of a body and a syphon tube and a vapor tube on the same side of the body. Dimensions of the extractor shall conform to the requirements of table I.

3.2.3 *Boiling flask.* The boiling flask shall be flat or round bottom, as specified (see 6.2), short neck type with a standard 24/40 outer, ground joint at the top. Capacity shall be 125 ml. for small, 250 ml. for medium. 300 ml. for large. An area for marking shall be provided by blasting or spot enamel. The area shall show no change or removal of enamel, if used, and shall be satisfactory for marking when tested as specified in 4.6.1.

TABLE I.—*Dimensions of extractor*

	Inside diameter of tube, mm. ± 3 mm.	Total length of body, mm. ± 8 mm.	Listed capacity to top of siphon, ml.	T joint	
				Top	Bottom
Small	30	185	50	34/45	24/40
Medium	40	205	85	45/50	24/40
Large	50	243	200	55/50	24/40

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3.3 Marking. All extraction assembly components shall be permanently and legibly marked with the name or registered trademark of the manufacturer.

3.4 Workmanship. Extraction assemblies shall be symmetrically made, properly annealed, and finished. All components shall be free from defects which impair their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Special provisions.

4.2.1 *Alternative inspection (including testing) procedures.* The supplier may utilize any alternative inspection procedure which will insure equal or better quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern.

4.2.2 *Supplemental evidence.* The supplier shall provide evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 have been satisfied.

4.3 Lotting. A lot shall consist of those extraction assemblies of the same size; or condensers, extractors, or boiling flasks of same size, made by one manufacturer and submitted for acceptance at one time.

4.4 Sampling.

4.4.1 *Nondestructive examination.* Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2 *For test.* A sample of three boiling flasks shall be taken at random from each lot of 100 flasks or fraction thereof. One third of the flasks shall be tested in accordance with 4.6.1.1, 4.6.1.2, and 4.6.1.3, respectively.

4.5 Examination. Sample extraction assemblies and level A and level B preparation for delivery requirements shall be examined in accordance with the classification of defects and with MIL-STD-105.

4.5.1 *Classification of defects.*

4.5.1.1 *Extraction assembly.*

Categories	Defects
Critical:	None defined
Major:	*AQL 4.0 percent defective*
101	Dimensions incorrect
102	Unsymmetrical in shape
103	Defective (broken, cracked, scratched)
104	Marking missing, illegible, incorrect

4.5.1.2 *Preparation for delivery (section 5).*

Categories	Defects
Critical:	None defined
Major:	*AQL 2.5 percent defective*
101	Unit, intermediate, or shipping container not as specified
102	Number of components per intermediate container not as specified
103	Unlike size components packaged per intermediate container
104	Closure of unit, intermediate, or shipping container not as specified
105	Unit, intermediate, or shipping container damaged
106	Cushioning or blocking of unit container not as specified
107	Marking of unit, intermediate, or shipping container illegible, incorrect, or incomplete

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4.6 Tests. Tests shall be conducted as follows:

4.6.1 *Area of marking permanency.*

4.6.1.1 *Alkali test.* Immerse the flask completely in normal sodium hydroxide at room temperature, cover, and boil for 1/2 hour, and let stand for 1 hour. Remove flask and rinse thoroughly with distilled water. Dry by rubbing vigorously with a cloth.

4.6.1.2 *Acid test (hydrochloric).* Immerse the flask completely in concentrated hydrochloric acid (sp gr 1.18) at room temperature, cover, and boil for 1/2 hour. Let stand for 1 hour. Remove flask and rinse thoroughly with distilled water. Dry by rubbing vigorously with a cloth.

4.6.1.3 *Acid test (sulfuric).* Immerse the flask completely in concentrated sulfuric acid (sp gr 1.84) at room temperature. Let stand 1 hour. Remove flask and rinse thoroughly with distilled water. Dry by rubbing vigorously with a cloth.

4.7 Acceptance/rejection criteria. If any flask fails to meet the test requirements of this specification, the lot represented shall be rejected.

5. PREPARATION FOR DELIVERY

For civil agency procurements, the definitions and applications of the levels of packaging and packing shall be in accordance with Fed. Std. No. 102.

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 *Level A.*

5.1.1.1 *Unit packaging.* Each condenser, each extractor, and each flask shall be packaged in an appropriate size fiber box conforming to type I, class 1, style RSC of PPP-B-636. Fiber blocking or sufficient cushioning material conforming to type II, class B, of PPP-C-843 shall be used in the boxes to prevent shifting and damage of contents. Boxes shall be closed in accordance with the appendix of the box specification.

5.1.1.2 *Intermediate packaging.*

5.1.1.2.1 *Extraction assembly.* When specified (see 6.2), one condenser, one extractor, and one flask of the size specified (see 6.2) unit packaged in accordance with 5.1.1.1 shall be intermediately packaged in a snug-fitting fiber box conforming to type I, class 1, style RSC of PPP-B-636. The box shall be closed in accordance with the appendix of the box specification.

5.1.1.2.2 *Condensers, extractors, and flasks.* The number of condensers, extractors, or flasks specified (see 6.2), unit packaged in accordance with 5.1.1.1 shall be intermediately packaged in a snug-fitting fiber box conforming to type I, class 1, style RSC of PPP-B-636. Each intermediate package shall contain only the type and size component specified (see 6.2). The box shall be closed in accordance with the appendix of the box specification.

5.1.2 *Level C.* Extraction assemblies or components, as specified (see 6.2) shall be packaged in accordance with the manufacturer's commercial practice suitable for shipment from the supply source to the receiving activity for immediate use.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 *Level A.* The number of extraction assemblies or the number of condensers, extractors, or flasks specified (see 6.2), intermediately packaged in accordance with 5.1.1.2, shall be packed in a box conforming to PPP-B-621, class 2, or PPP-B-601, overseas type. As far as practical, exterior containers shall be of uniform size and shape, be of minimum cube and tare, and contain identical quantities of like items. Gross weight shall not exceed the weight limitations of the ap-

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plicable box specification or appendix thereto. Containers shall be closed and strapped in accordance with the applicable box specification or appendix thereto.

5.2.2 *Level B.* Extraction assemblies or components, as specified (see 6.2), shall be packed in accordance with 5.2.1 except that shipping containers shall conform to PPP-B-621, class 1; PPP-B-601, domestic type; or PPP-B-636, type I, class 1, fabricated from not less than 275-pound test fiberboard.

5.2.3 *Level C.* Extraction assemblies or components, as specified (see 6.2), shall be packed to insure carrier acceptance and safe delivery at first domestic destination. Containers shall be in accordance with Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation.

5.3 **Marking.**

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or order, marking shall be in accordance with Fed. Std. No. 123.

5.3.2 *Military agencies.* In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 **Intended use.** This specification covers one extraction assembly intended to be used for the laboratory determination of the constituents in such substances as fats, oils, gums, etc., which are soluble in volatile liquid.

6.2 **Ordering data.** Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Level of packaging and packing required.
(*c*) Size of assembly or type and size of component desired.
(*d*) The number of condensers, extractors, or flasks per intermediate package and shipping container required.

6.3 **Military use.** Items procured under this specification for military use are to be limited to the variety shown on the applicable military standard or military supply standards. Personnel of the military departments are requested to refer to these documents for guidance.

6.4 **Transportation description.** Transportation description applicable to these items is:

Glassware, chemical laboratory.
Carload minimum weight 18,000 pounds, subject to Rule 34, Uniform Freight Classification.
Glassware, laboratory.
Motor volume minimum weight 18,000 pounds.

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Copies of this specification may be purchased for 5 cents each.

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